

Calcularis Data Analysis Report

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I. INTRODUCTION

The purpose of this report is to provide a comprehensive analysis of the user data collected from the calcularis dataset. The dataset serves as a valuable resource containing a wealth of information regarding the performance of students who utilize the company's mathematics fun exercises. These exercises are designed to enhance students' skills in a wide range of mathematical topics, including counting, division, multiplication, and more. By analyzing this dataset, we aim to gain insights into the effectiveness of the calcularis program and explore predictive models for user performance.

Research Questions:

The analysis primarily focuses on addressing two crucial questions that can significantly impact the educational outcomes of calcularis users:

- 1) **Performance Prediction:** The first question focuses on developing predictive models that effectively forecast user performance across a set of skills. Specifically, we aim to answer the question of how we can predict user performance across these skills. By leveraging the rich dataset at our disposal, we can explore various factors and patterns that contribute to a student's performance. The development of such predictive capabilities holds immense value as it enables us to tailor the calcularis program to meet the unique needs of individual students, providing them with personalized recommendations to enhance their learning journey. Additionally, this analysis allows us to visualize the learning curve of each student, represented by their user id feature.
- 2) **Performance Comparison:** We investigate the impact of the free training feature offered by calcularis on student performance, providing insights into the effectiveness of this training method. However, it is important to acknowledge the potential imbalanced aspect of the data. Therefore, we aim to analyze and understand whether a notable difference exists in the performance between students who utilize the free training offered by calcularis and those who do not. To address this issue, we will perform propensity score matching and approach the analysis from an observational study perspective. By comparing the outcomes of these two groups, we can determine whether the availability of free training positively or negatively influences student performance. This comparison provide valuable insights into the impact of the free training feature and its role in shaping student achievement.

In the following sections, we will describe the methodology used for the analysis, present the results, and discuss the implications of the findings.

II. METHODOLOGY

A. Data Description and Exploratory Analysis

In this section, we describe the datasets used in the analysis and the preprocessing steps performed to clean and transform the data.

1) **Datasets:** The provided datasets are divided into three main tables: *users*, *events*, and *subtasks* dataframes. These tables contain information about the users, their interactions, and the subtasks they have performed in the calcularis program.

2) **Preprocessing:** The following preprocessing steps were applied to the datasets:

- 1) **Handling Missing Values:** The events and user tables were examined for missing values, and any rows with missing values were removed. After this step, the dataset consisted of 2,153,208 events performed by 48,778 distinct users.
- 2) **Filtering Valid Subtasks:** In the events table, the *subtask* feature contains dictionaries with various subtask details. Some subtask entries lack the *correctAnswerObject* variable, making it impossible to assess the student's proximity to the correct answer. To filter out valid subtasks, we considered a subtask valid if the *hasProperResult* feature in the dictionary is set to True, and all of the *correct*, *type*, *answer*, and *correctAnswerObject* fields are present and valid. Non-valid subtask entries were filtered out, resulting in 2,124,767 remaining event entries.
- 3) **Computing Score Metric:** To account for the variable difficulty of questions, a score metric was computed. We assigned weights to different difficulty ranges, such as {'R10': 1, 'R20': 2, 'R100': 10, 'R1000': 100}. The score for a given skill and difficulty range is computed as the sum of the earned points divided by the maximum possible score.
- 4) **Mapping Skill IDs to Domains:** To understand the skill associated with each event, we mapped the skill ID field to the corresponding domain. The following steps were performed:
 - a) The skill names were preprocessed and tokenized.
 - b) Clustering was performed on the preprocessed skill names using K-means with Jaccard set distance.
 - c) The clustering resulted in six well-separated clusters: ('Addition', 'Division', 'Multiplication', 'Number_line', 'Subtraction').
 - d) Two clusters had different naming semantics but represented the same skill, namely 'Number line' (as suggested from Calcularis spider chart skill categories).
 - e) The two skill clusters were merged under the skill set 'Number line'.
- 5) **Filtering Events for Specific Skills:** we made sure that each event only references one of the following skills: ('Addition', 'Division', 'Multiplication', 'Number_line', 'Subtraction').
- 6) **Filtering Users with Sufficient Events:** Users who did not perform enough events (less than 5 games) were filtered out to avoid bias in the analysis. This resulted in a subset of events where users played more than 5 games, with a length of 1,500,753, namely events-pre.
- 7) **Exploding Subtask Field:** The subtask field in the events table, which is a dictionary, was exploded for each user, resulting in a separate entry for each subtask performed, namely table sutasks-pre.
- 8) **Feature Selection and Computation:** Finally, we create a new dataframe with all the processed data indexed by user_id, named users-pre. Only features of interest were retained, including learning_time of each user, locality of each user, the number of events (represented as games) per user, the number of answers per user, the absolute percentage of correct answers

per user, weighted scores for each category and each difficulty, the maximum level of difficulty reached for each skill, and the free training time usage in milliseconds.

Additionally, distributions of tasks were plotted based on skill names and difficulty ranges (e.g., R1000, R100, R20, R10).

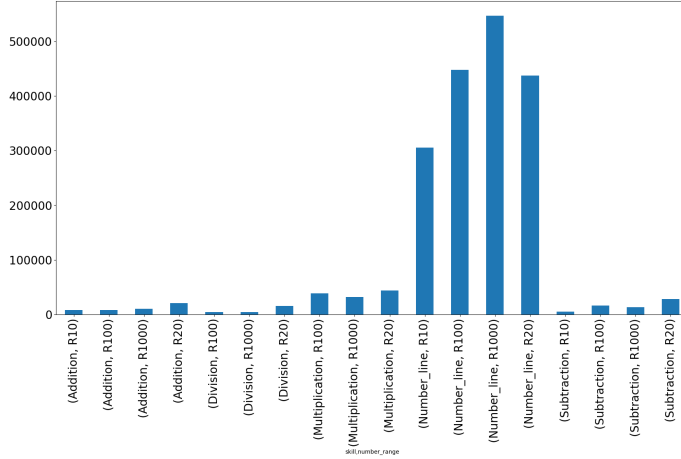


Fig. 1. distributions of tasks given a skill name and the difficulty range.

The analysis of figure 1 revealed that the most popular skills were related to Number line skills, which is understandable as the majority of the games require mathematical number representations and involve beginner-level math knowledge about numbers.

The above preprocessing steps and distribution analysis provide a solid foundation for further exploration and analysis of the calculator dataset, enabling us to gain insights into user performance and the impact of free training on student achievement.

Additionally, as part of the analysis, we included a visualization that allowed us to assess user strengths and weaknesses. This visualization took the form of a spider plot, which provided an overview of a user's proficiency across different skills. By plotting the scores or performance metrics for each skill on the spider plot, we were able to identify areas where a user excelled and areas that required improvement.

This spider plot in figure 2 visualization provided a comprehensive and intuitive representation of a user's skill profile

B. Data Analysis Approach :

1) *Predictive Modeling*: In the process of implementing the predictive model used to forecast user performance on a set of skills, the goal was to develop models that leverage previous interactions with educational exercises to predict the performance of students. Specifically, we employed a multi-learn Bayesian knowledge tracing model, which was trained on different tasks based on the nature of the skill they convey and the difficulty range associated with each task. For this purpose, we followed the following steps :

• Knowledge Tracing Model :

We utilized a knowledge tracing model, specifically the Bayesian Knowledge Tracing (BKT) algorithm. The BKT model was trained on user data for a clustered set of tasks that corresponded to the same general skill, such as Addition, Subtraction, or other mathematical concepts. The models were designed to consider the possibility of users forgetting a skill over time, which is a natural occurrence in the learning process. By incorporating this aspect into the model, we aimed to capture the dynamics of skill

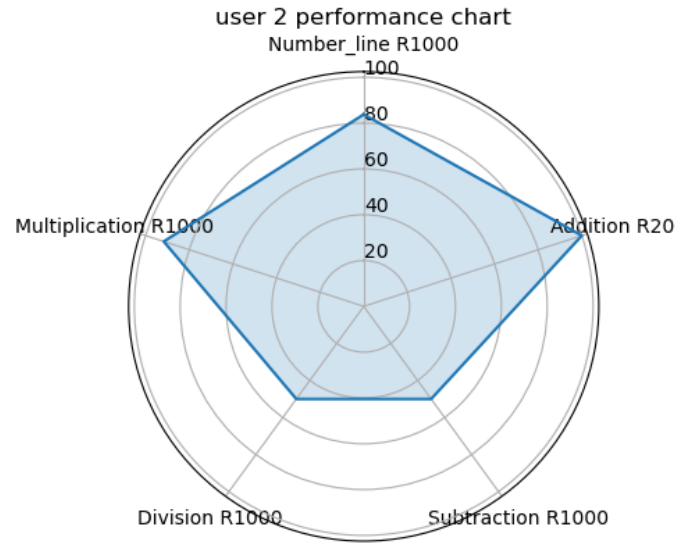


Fig. 2. visualisation of user 2 performances across the skills.

retention and decay to make more accurate predictions of user performance.

The performance of the models was evaluated using appropriate metrics. The Root Mean Square Error (RMSE) was employed as a measure of accuracy, quantifying the differences between the predicted and actual performance values. Additionally, the Area Under the Curve (AUC) was used as a precision metric, assessing the model's ability to correctly predict the performance of students.

• Data Preprocessing for BKT :

Prior to model training, the dataset underwent preprocessing steps to ensure it was suitable for knowledge tracing. Specifically, we encoded student interactions and their corresponding skill outcomes into a format compatible with the knowledge tracing model. This preprocessing step prepared the dataset for the subsequent training and evaluation stages. In the following we list some major preprocessing steps that we took into consideration:

- Preparing a multipurpose subtasks dataframe that also includes categorical skill names.
- Categorization of the skill names as outputted by our previous clustering model 4
- Preparing specific time series dataframes to split data into training and testing datasets, 80% of the first weeks as training set and 20% for the last weeks as testing set. The split is with respect to every user.
- We also drop all user that performed under 50 sub-tasks, This threshold was taken since the average completed sub-tasks per user is ≈ 57 as shown by figure 3

• **Model Building for BKT :** We considered two different approaches when it came to building our BKT models. Both models were configured with the same default prior-probabilities such as prior-learn, prior-forget, which are determined automatically by the pyBKModel library ¹

- a) **Multi-learn on users:** In this model, the *multilearn* parameter was set on users. Therefore the model attempts

¹<https://pypi.org/project/pyBKT/1.0.1/>

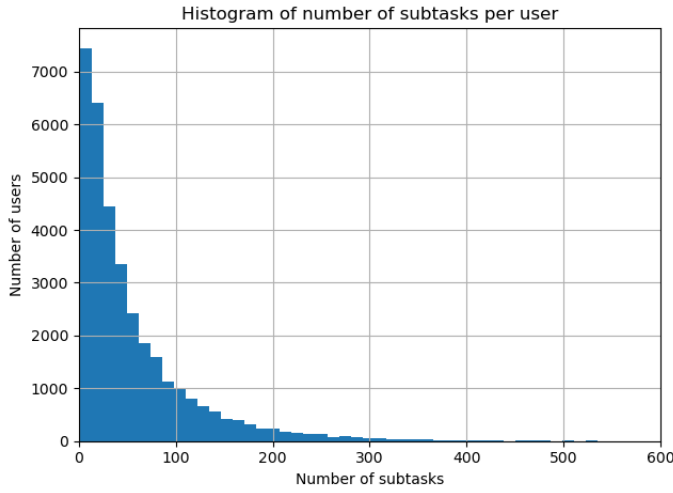


Fig. 3. Visualisation of sub-tasks frequencies distribution

to capture the probability that a user has mastered skill category using the preprocessed training data. It then outputs the *learn* probability of user for a specific skill.

- b) **Time series multi-learn:** Alternatively, in this model we make use of the time series dataset for each skill category given a range of difficulty. This was considered in order to capture the user's learning probability with respect to the first weeks time frame. We then tested the model's performance with the latest time frame weeks.

- **Model Evaluation :**

After training the knowledge tracing model on the preprocessed dataset, we evaluated its performance using appropriate metrics. The evaluation focused on assessing the accuracy and precision of the model's predictions. These chosen metrics provided insights into the model's effectiveness in predicting user performance across different skills.

2) *Inferential analysis:* In this subsection, we describe the inferential analysis approach used to examine the effect of utilizing free training offered by calcaris on user performance. The steps followed in the analysis are outlined below:

- **Defining Treatment and Feature Selection:** The treatment variable is defined as the `total_free_training_time_ms` feature. All relevant features, including `ui_locale`, `number_of_games`, and `learning_time_ms`, along with the performance scores of users, are selected for analysis. The categorical feature `ui_locale` is one-hot encoded to facilitate further analysis.
- **Propensity Score Computation:** Propensity scores are computed using logistic regression. The logistic regression model predicts the probabilities of users utilizing the free training. The predicted probabilities are then transformed into logit form. The computed propensity scores are added to the DataFrame, allowing for further analysis.
- **Density Distribution Plots:** Density distribution plots are generated for both the propensity scores and logit of propensity scores. The density distribution in figure 4 of logit propensity scores provides insights into the balance between treated and untreated users, indicating the potential imbalance in the data. The overlapping populations on the plot suggest the need for

Density distribution plots for propensity score and logit(propensity score).

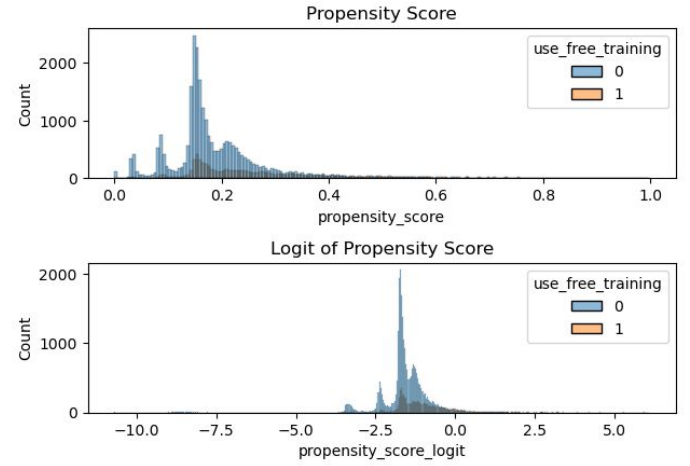


Fig. 4.

matching.

- **Threshold Selection:** A K-nearest neighbors (KNN) algorithm is set up using the chosen threshold value, namely 25% of the standard deviation of the propensity scores, to find the nearest matches for each user. This threshold value helps define the acceptable range for matching. In other words, we want to find control users whose propensity scores are close to the treated user's propensity score, within this specified range.
- **Matching Process:** The matching process aims to create pairs of treated and control users based on their propensity scores. It involves iterating over the dataset and finding the nearest neighbors in the control group for each treated user. The matching is performed based on predefined criteria, ensuring that the matched pairs are close in terms of their propensity scores. (see algorithm 1 describing this procedure).
- **Matched Observations:** The total number of matched observations in the control and treatment groups is obtained. This information provides insights into the matched sample sizes for further analysis. The total observations in control is 26716 users, the total observations in treatment is 7160 users and finally the total matched observations is 6535 users. In addition, several checks and validations are performed, including checking the uniqueness of user IDs in the control and treatment groups.
- **Analyzing the Free Training Effect on Scores:** The effect of free training on the `absolute_percentage_correct` feature is analyzed by calculating the mean and standard deviation of scores separately for the control and treatment groups. Please refer to Table 1 to obtain the results.

The difference between the control and treatment groups is plotted in figure 5, and the mean difference is compared. A t-test is performed to determine the significance of the differences between the two groups. We obtain a p-value of 0, which indicates that the differences are statistically significant, allowing us to reject the null hypothesis. Furthermore, we plotted the differences between the 2 groups for each specific skill : addition, multiplication, etc. (refer to appendix for the plots). Although the plots indicate a dominance of users who do not utilize free training in most skills, our findings suggest that users who do utilize free training tend to perform better. This observation can be attributed to not only the higher absolute

Algorithm 1: Matching Process Algorithm

Data: *df*: Data-Frame containing the data, *treatment*: Name of the treatment feature

Result: *matched_control*: Set of indices representing the matched control users

Algorithm:

```
Function MatchingProcess(df, treatment):  
    Initialize an empty set matched_control;  
    foreach current_index, row in df do  
        match only if the user is part of treatment;  
        if row[treatment] == 1 then  
            Iterate over the nearest neighbor indexes except  
            self;  
            foreach idx do  
                if (df.loc[idx][treatment] == 0) and (idx not  
                in matched_control) then  
                    df.loc[current_index, 'matched'] = idx;  
                    Add idx to matched_control set;  
                    Break the inner loop;  
                end  
            end  
        end  
    end  
    return matched_control;
```

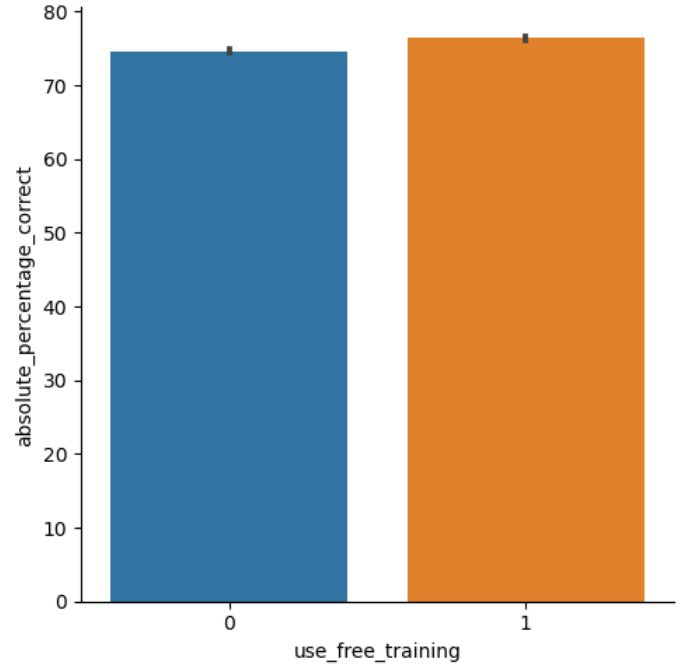


Fig. 5. comparison of the free_training vs no_free_training users in their absolute correct percentage score. The 1 represents our treatment group those who use free training and 0 are the control group.

TABLE I
FREE TRAINING EFFECT ON SCORES

use_free_training	Mean	Standard Deviation
0	74.59995	13.04115
1	76.44859	11.86745

average correct score feature previously plotted, but also to the fact that the 'Number_line' skill category is more prevalent in our dataset compared to other skills. As a result, the higher performance of free training users may be influenced by the abundance of data points related to the 'Number_line' skill.

III. EXPERIMENTAL EVALUATION

1) *Evaluation of Predictive BKT Model:* The performance of the predictive BKT model was evaluated using two metrics: Root Mean Squared Error (RMSE) for accuracy and Area Under the Curve (AUC) for precision. Additionally, the model relied on the predictions of grasped skills to compute Mean Squared Error (MSE) per user.

1) First BKT Model

As shown in Figure 6, the first BKT model achieved a training precision of approximately 70% (AUC) and an accuracy of around 40% (RMSE) per difficulty level.

From Figure 7, we observe a similar training precision and accuracy per skill domain. The testing evaluation shows comparable performance to the training, indicating that the model has generalized well from the training set to the unseen testing set. This suggests that the model is not overfitting.

2) Second BKT Model

Figure 8 demonstrates that the second BKT model achieved a training precision of approximately 70% (AUC) and an accuracy of around 40% (RMSE) per skill domain.

Train-Test Split Evaluation

For the train-test split evaluation, 80% of the data was used for training, while the remaining 20% was utilized for testing. To account for data imbalance across users, two episodes were conducted. The first episode considered a threshold of 50 subtasks performed, while the second episode used a threshold of 100 subtasks.

The testing evaluation plot depicted in Figure 9 presents the results obtained with a threshold of 50 subtasks. In this evaluation, the precision is observed to be lower compared to the training results. This discrepancy suggests that the training data may not accurately represent the entire dataset or may lack diversity in terms of the variations and patterns present in the data. The discrepancy can be attributed to the fact that the remaining 20% of the data was selected as a time series sample. As a result, users in the testing set may have encountered different skills compared to the ones encountered in the training set.

The testing evaluation plot with a threshold of 100 subtasks, as shown in Figure 10, reveals better results for the Number Line category. However, the Multiplication skill does not show the same improvement. This discrepancy can be attributed to the fact that the Number Line category is more prevalent in the dataset, leading to a higher threshold of subtasks performed. On the other hand, the Multiplication skill may be underrepresented in the data, resulting in lower performance.

Overall, the evaluation of the predictive BKT model provides insights into its precision, accuracy, and generalization. The findings highlight the strengths of the model and areas that may require further improvement, such as addressing imbalances in the training data and enhancing the representation of certain skills.

Training Evaluation of skill_id BKT model

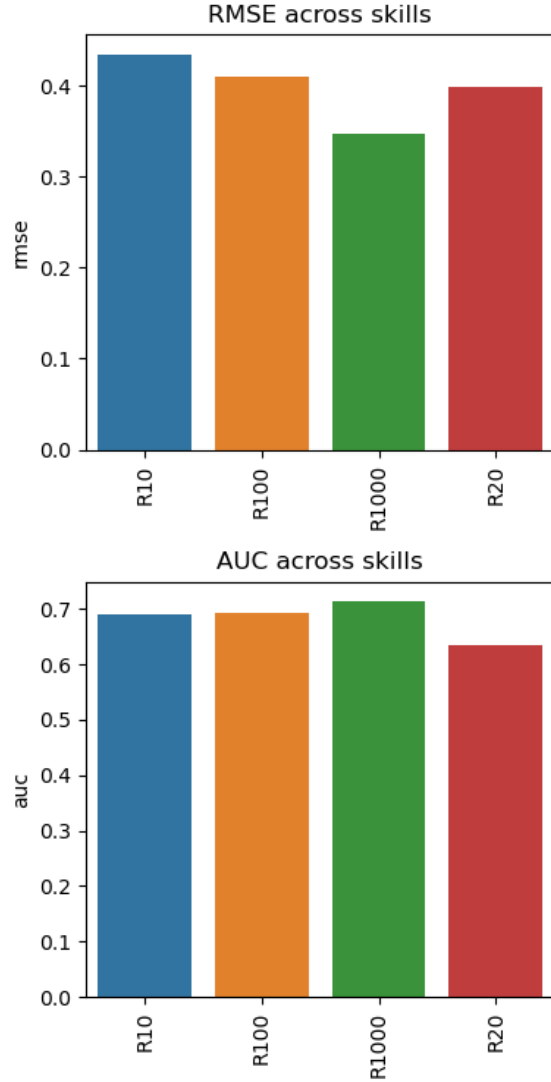


Fig. 6. Training evaluation plot for the first BKT model

2) *inferential analysis evaluation*: In the initial phase of our inferential evaluation analysis, we compare the densities of propensity score logits before and after performing the matching process. This comparison allowed us to assess the change in densities and evaluate the effectiveness of the matching process.

Figure 11 visually presents the densities of propensity score logits before and after matching. From the figure, we can observe that the densities after matching exhibit a comparable shape. The overlapping and similar distributions indicate that the matching process successfully aligned the propensity score distributions of the two groups.

As a result, any differences observed between the groups can be more confidently attributed to the treatment effect, as potential confounding factors have been minimized.

Furthermore, in the context of our observational study, we utilize Cohen's d function to assess the magnitude of the performance difference between the two groups under comparison: users who utilized free training and those who did not. Cohen's d is a statistical measure that quantifies the effect size, indicating the extent of the

80-20 Train-Test split Evaluation of skill_id BKT model

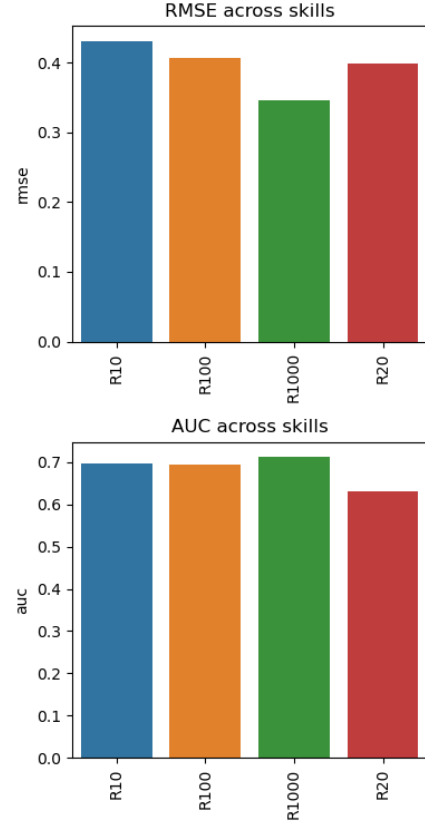


Fig. 7. Testing evaluation plot for the first BKT model

difference between the groups. A larger Cohen's d value signifies a larger difference and, consequently, a stronger effect size.

Typically, Cohen's d values of 0.2, 0.5, and 0.8 or higher are regarded as representing small, medium, and large effect sizes, respectively. By calculating Cohen's d before and after matching the users, we aimed to evaluate the impact of the matching process on reducing the effect size. The latter can be used to determine the magnitude of the difference in performance between two groups being compared (free training vs no free training in our case).

As shown in Figure 12, we can observe the reduction in the effect size after performing the matching of users. The figure visually represents the change in the effect size, indicating a considerable decrease.

By reducing the effect size through matching, we enhance the comparability between the free training and no free training groups, making the results more reliable and valid. The decreased effect size suggests that the matching process successfully addressed potential confounding factors, resulting in a more balanced and fair comparison between the two groups.

IV. DISCUSSION AND IMPLICATIONS

Overall, our experimental methods held satisfying results. The Bkt models achieved overall good accuracies as shown in performed prediction plots on unseen data 17 18.

Since the skill-id model achieves similar accuracy on both the training and testing data 13 14, it suggests that the model has generalized well from the training set to the unseen testing set. This means that the

Training Evaluation of user_id BKT model (threshold=50)

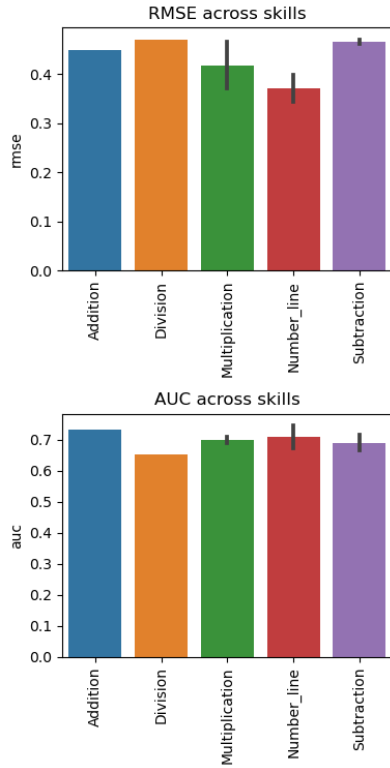


Fig. 8. Training evaluation plot for the second BKT model

model is not overfitting, and seem to perform well on new, unseen data. It also implies that it trained on sufficient amount of data to capture a patterns and variations and that the model's complexity is appropriate for the task at hand.

Overall, this model was less affected by the threshold of subtasks performed, and seems to be robust to data imbalance since it is more general and tries to grasp how an average student would learn a given skill in given difficulty.

On the other hand, the user-id model is more specific and tries to grasp how a does a specific user learn in a specific skill and difficulty. Its decrease on testing accuracy 15 16 suggests that the distribution of the testing data is significantly different in comparison to the training data. This can be explained by the fact that the data was organized as a time series table. While a student might target specific skills in his first weeks of usage, he might tackle completely different skills later on.

However it achieved higher best accuracies in some skills, namely Division R1000, even if the average accuracy was significantly below the first model. This could hint to fine tune a model for each user, which would be costly in terms of time and resources.

Regarding the Observational Study, our results held a p-value of 0, indicating that the observed difference in means between the two groups is extremely unlikely to have occurred by chance alone. It provides strong evidence to reject the null hypothesis, which states that there is no difference between free training and solely guided training users. Along with the p-value, we considered the magnitude of the observed differences between the means with the effect size, which held small effects using Cohen's d function.

One important limitation of our study is that the matching process did

80-20 Train-Test split Evaluation of user_id BKT model (threshold=50)

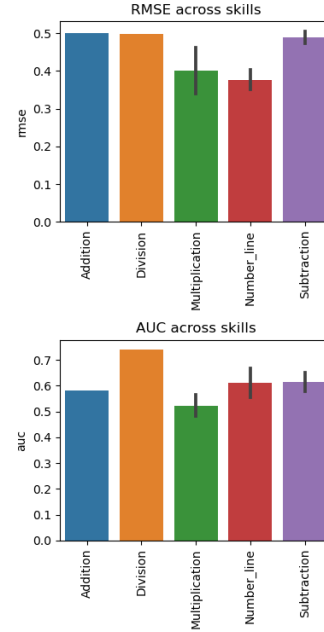


Fig. 9. Testing evaluation plot with a threshold of 50 subtasks

80-20 Train-Test split Evaluation of user_id BKT model (threshold=100)

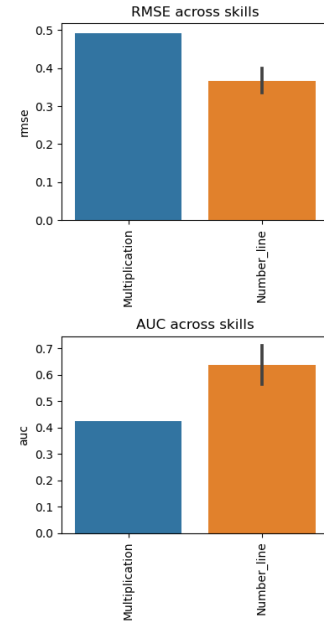


Fig. 10. Testing evaluation plot with a threshold of 100 subtasks

not take into account the level of users in different skill categories. This could be seen as unfair since the level of skill can significantly influence the outcome being measured. Although we recognized the importance of considering skill level during the matching process, we faced challenges in finding a suitable approach that would avoid introducing bias into our study. Balancing skill levels across groups could potentially lead to unintentional biases, such as restricting the range of skill levels or inadvertently selecting individuals who are not representative of the population.

Comparison of propensity_score_logit split by treatment status.

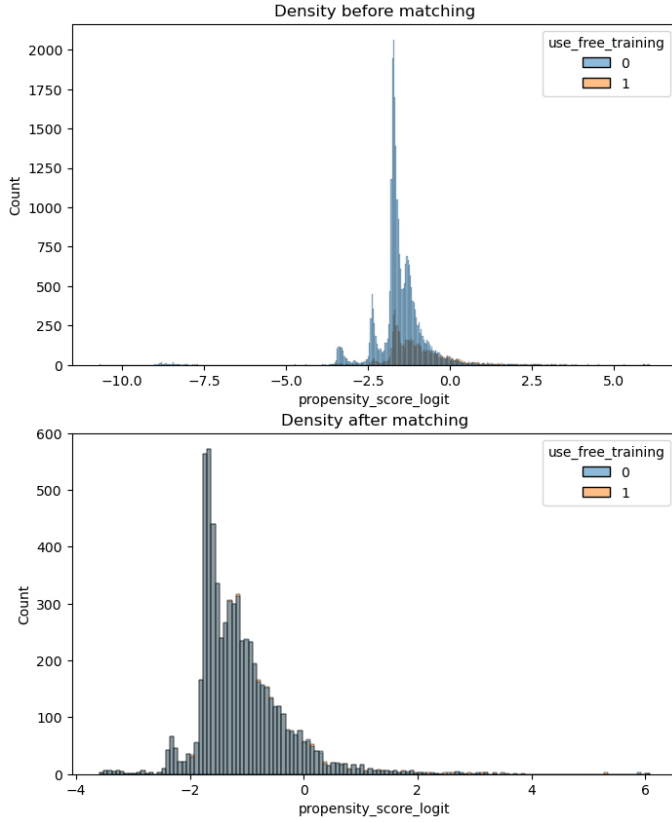


Fig. 11.

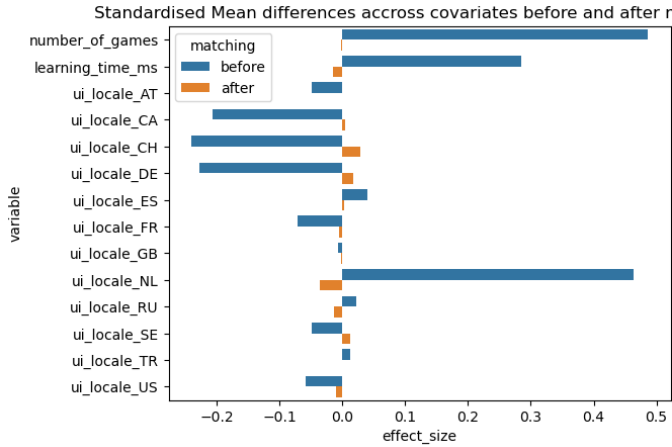


Fig. 12.

However, we acknowledge that further reflection and is needed to address this limitation in future research. Alternatively, stratifying the analysis based on skill levels or conducting subgroup analyses could provide insights into how the observed differences in means may vary across different skill levels.

		model	auc	rmse
skill_name	number_range			
id\model	R10	Model(parallel=True, num_fits=5, seed=0, defau...	0.69054	0.43444
	R100	Model(parallel=True, num_fits=5, seed=0, defau...	0.69363	0.40933
	R1000	Model(parallel=True, num_fits=5, seed=0, defau...	0.71324	0.34709
	R20	Model(parallel=True, num_fits=5, seed=0, defau...	0.63451	0.39841

Fig. 13. Table with all accuracies obtained for training evaluation for the first model

		model	auc	rmse
skill_name	number_range			
id\model	R10	Model(parallel=True, num_fits=5, seed=0, defau...	0.69757	0.43110
	R100	Model(parallel=True, num_fits=5, seed=0, defau...	0.69544	0.40675
	R1000	Model(parallel=True, num_fits=5, seed=0, defau...	0.71204	0.34653
	R20	Model(parallel=True, num_fits=5, seed=0, defau...	0.63174	0.39950

Fig. 14. Table with all accuracies obtained for testing evaluation for the first model

V. CONCLUSION

In conclusion, the analysis of the calcaris dataset provided valuable insights into user performance and the impact of free training on student achievement. The analysis addressed two main research questions: performance prediction and performance comparison.

Regarding performance prediction, a Bayesian Knowledge Tracing (BKT) model was implemented to forecast user performance across different skills. The evaluation of the BKT models revealed a training precision of approximately 70% (AUC) and an accuracy of around 40% (RMSE) per difficulty level. The testing evaluation demonstrated comparable performance to the training, indicating good generalization of the models. This suggests that the BKT models can effectively predict user performance and tailor the calcaris program to meet individual student needs.

Regarding performance comparison, an inferential analysis was conducted to examine the impact of utilizing free training on user performance. Propensity score matching was performed to address potential confounding factors. The analysis revealed a statistically significant difference in performance between users who utilized free training and those who did not. The users who utilized free training showed higher scores and better performance overall. This finding highlights the effectiveness of the free training feature offered by calcaris in enhancing student achievement.

The results of the analysis have significant implications for educational interventions and personalized learning. The predictive BKT models can inform the development of personalized recommendations for students, allowing for targeted support and tailored interventions. The impact of free training on user performance emphasizes the importance of providing additional resources and opportunities for students to enhance their skills and knowledge.

Further research and exploration of additional variables and populations would contribute to a deeper understanding of the factors influencing student performance and improve the effectiveness of

skill_name	number_range	model	auc	rmse
Addition	R1000	Model(parallel=True, num_fits=5, seed=0, defau...	0.73271	0.45011
Division	R1000	Model(parallel=True, num_fits=5, seed=0, defau...	0.65091	0.47030
Multiplication	R100	Model(parallel=True, num_fits=5, seed=0, defau...	0.68961	0.46674
	R1000	Model(parallel=True, num_fits=5, seed=0, defau...	0.70743	0.37045
	R10	Model(parallel=True, num_fits=5, seed=0, defau...	0.76942	0.36873
Number_line	R100	Model(parallel=True, num_fits=5, seed=0, defau...	0.69435	0.40668
	R1000	Model(parallel=True, num_fits=5, seed=0, defau...	0.70063	0.33092
	R20	Model(parallel=True, num_fits=5, seed=0, defau...	0.66939	0.38231
Subtraction	R100	Model(parallel=True, num_fits=5, seed=0, defau...	0.70483	0.46174
	R1000	Model(parallel=True, num_fits=5, seed=0, defau...	0.65521	0.47331
	R20	Model(parallel=True, num_fits=5, seed=0, defau...	0.71097	0.46348

Fig. 15. Table with all accuracies obtained for training evaluation for the second model

skill_name	number_range	model	auc	rmse
Addition	R1000	Model(parallel=True, num_fits=5, seed=0, defau...	0.58214	0.50161
Division	R1000	Model(parallel=True, num_fits=5, seed=0, defau...	0.74013	0.49748
Multiplication	R100	Model(parallel=True, num_fits=5, seed=0, defau...	0.56357	0.46320
	R1000	Model(parallel=True, num_fits=5, seed=0, defau...	0.48060	0.34030
	R10	Model(parallel=True, num_fits=5, seed=0, defau...	0.70772	0.36724
Number_line	R100	Model(parallel=True, num_fits=5, seed=0, defau...	0.58798	0.41337
	R1000	Model(parallel=True, num_fits=5, seed=0, defau...	0.58515	0.34171
	R20	Model(parallel=True, num_fits=5, seed=0, defau...	0.56236	0.38392
Subtraction	R100	Model(parallel=True, num_fits=5, seed=0, defau...	0.57327	0.51314
	R1000	Model(parallel=True, num_fits=5, seed=0, defau...	0.60963	0.47918
	R20	Model(parallel=True, num_fits=5, seed=0, defau...	0.66332	0.47598

Fig. 16. Table with all accuracies obtained for testing evaluation with threshold 50 for the second model

educational interventions. By continuously analyzing and evaluating the data, calcaris can further refine its program and provide a more personalized and impactful learning experience for students.

Overall, the analysis of the calcaris dataset provides valuable insights into user performance and the effectiveness of educational interventions. These insights can inform educational practices and contribute to the continuous improvement of the calcaris program.

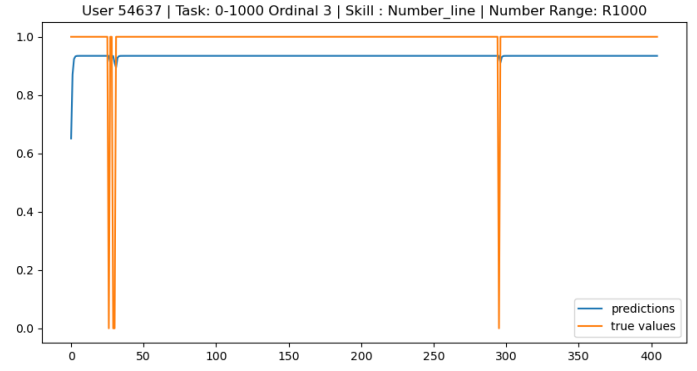


Fig. 17. Prediction of testing data of a specific user on a specific skill using the user-id BKT model

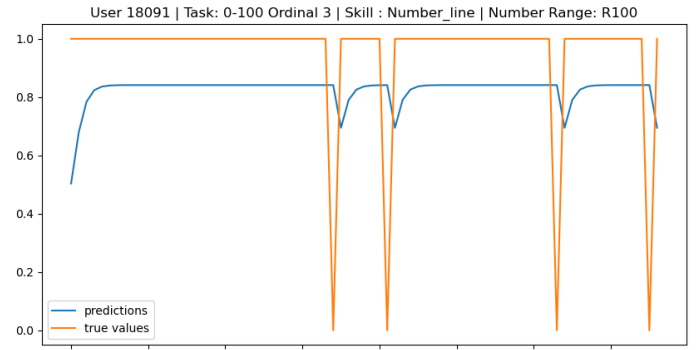


Fig. 18. Prediction of testing data of a specific user on a specific skill using the skill-id BKT model

APPENDIX

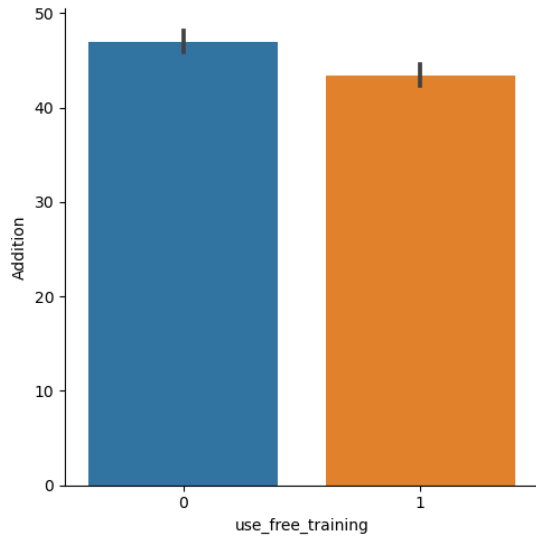


Fig. 19. comparison of the free training vs no free training users in their

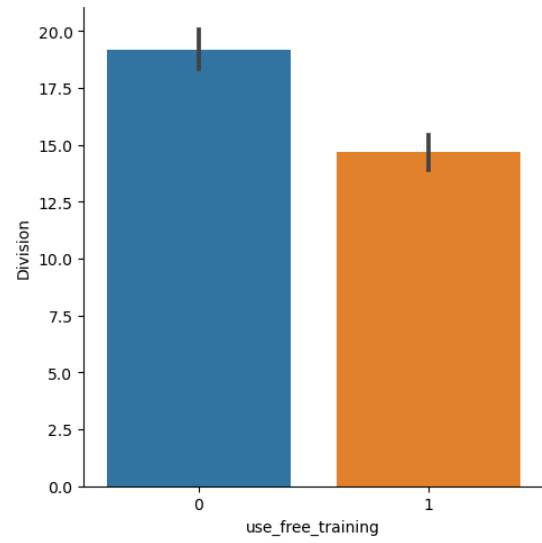


Fig. 21. comparison of the free training vs no free training users in

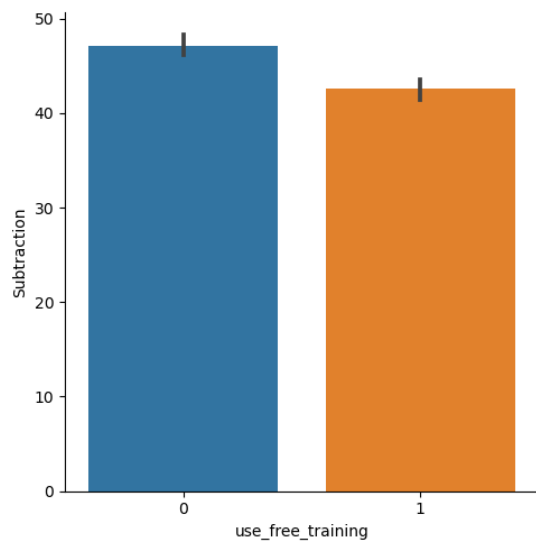


Fig. 20. comparison of the free training vs no free training users in their

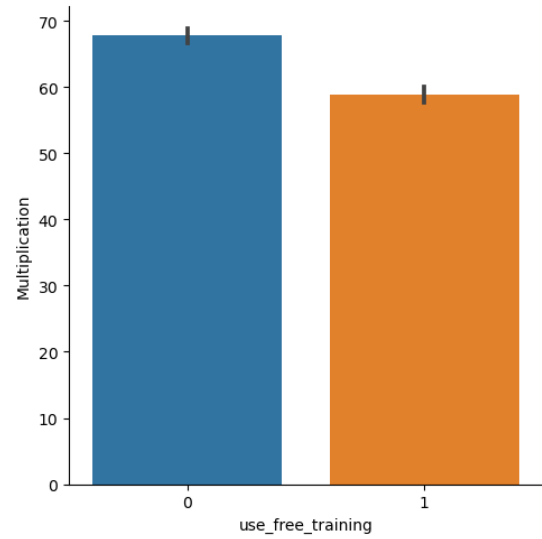


Fig. 22. comparison of the free training vs no free training users in their

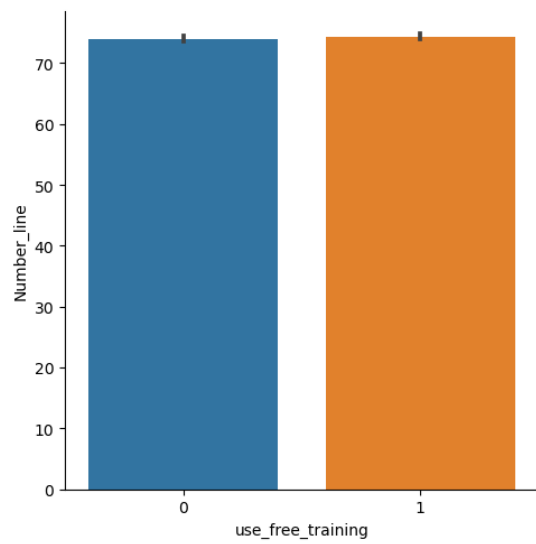


Fig. 23. comparison of the free training vs no free training users in their